

Chapter 18

AI in the Evolving Workplace



Peiqi Guo and Philip Young P. Hong

1 Labor Market Disruption in the Age of AI

In this section, we examine the potential effects of artificial intelligence (AI) on labor market disruption. We begin by identifying the underlying causes of labor market disruption and reviewing historical cases to better understand its motivations. We then explore how AI contributes to, or even triggers, such disruption. This section will also highlight current trends and future projections to outline AI's broader impact on the labor market. In addition, we introduce the concepts of job polarization and labor market segmentation. Finally, a set of case studies will be presented to further illustrate and contextualize the key issues discussed above.

1.1 Trends in AI-Driven Automation

1.1.1 Historical Context of Technological Disruption

Disruption in the labor market is neither rare nor unusual. It has consistently accompanied human history, often driven by technological development and evolving social and economic structures. The Industrial Revolution provides a powerful example to illustrate how technological advances have repeatedly caused significant shifts in labor systems. The term “industrial revolution” has been defined as a “step

P. Guo (✉)

School of Public Health, The University of Texas Health Science Center at Houston, Houston, TX, USA

e-mail: peiqi.guo@uth.tmc.edu

P. Y. P. Hong

School of Social Work, University of Georgia, Athens, GA, USA

e-mail: pyphong@uga.edu

into production technology which was completely different from the past” (Groumpos, 2021).

To date, three industrial revolutions have occurred. The First Industrial Revolution began in the early 1800s, the Second in the late 1800s through the early 1900s, and the Third happened in the late 1900s to early 2000s. As most scholars agree, the First Industrial Revolution originated in Great Britain and was marked by the transition from hand production to machinery. Labor-intensive sectors such as the textile industry were transformed by mechanized power sources, including steam and water power. Large-scale production systems began to emerge, exemplified by mechanized factories. This revolution catalyzed the transition from an agrarian to an industrial economy. Rural populations migrated to urban centers in response to factory labor demands and economic incentives. These migrations and subsequent demographic changes began reshaping the labor market.

The Second Industrial Revolution was characterized by the rise of electricity and the invention and implementation of the assembly line. Some scholars refer to this era as the “Technological Revolution.” Electricity became increasingly integrated into daily life, and inventions such as the telephone, electric light bulb, and radio radically reshaped society and accelerated urbanization. As industrialization progressed, labor demand continued to grow, further transforming the labor market.

The Third Industrial Revolution—also known as the Automation Revolution—emerged in the 1950s. It was defined by the use of programmable automation and the introduction of computers, which enabled fully automated production processes. This period also gave rise to foundational digital technologies, including semiconductors, personal computers, smartphones, and the Internet. As the world transitioned into the digital era, labor markets were disrupted and restructured once again. New occupations, such as software engineering, were created. Existing occupations were redefined or diminished, depending on their compatibility with digital workflows.

The Fourth Industrial Revolution is currently unfolding—or may already be well underway. This phase is rooted in digital technologies and is reshaping nearly every aspect of human society. Key innovations include the Internet of Things (IoT), advanced robotics, and perhaps most significantly, artificial intelligence (AI). A growing body of evidence suggests that the development of artificial general intelligence (AGI)—systems possessing human-comparable intelligence—may be on the horizon (McLean et al., 2023). The potential adoption and integration of AGI into daily life remains a subject of considerable debate. Proponents argue that AGI can significantly improve productivity and accelerate scientific progress. However, critics express concern over the social risks it may introduce, including mass unemployment and safety threats to societal stability (Nguyen & Vo, 2022).

As the historical overview above suggests, labor market disruptions are frequently associated with industrial revolutions. At the heart of such disruptions are the emergence and adoption of disruptive technologies. These technologies typically share several defining characteristics: (1) they are innovative and fundamentally transformative; (2) they can be implemented on a large scale; and (3) they

possess the capacity to restructure existing workflows. Technologies such as steam power, mechanization, electricity, and automation have all exemplified these traits.

Artificial intelligence also aligns with these characteristics. First, AI is an emerging disruptive technology with the potential to radically alter existing industries—for example, by replacing certain categories of human labor. Second, supporting technologies such as cloud computing and edge deployment have made large-scale AI implementation technically feasible. The widespread adoption of applications like ChatGPT further illustrates its scalability. Third, AI can reshape workflows. Just as human labor was reduced through mechanization and assembly-line production, AI can further displace human involvement in many processes. One key difference between AI and past industrial revolution technologies is the pace of transformation: AI has the potential to bring about widespread change in a significantly shorter timeframe.

1.1.2 Definition and Scope of AI in the Labor Market

We have previously discussed artificial intelligence in earlier sections, but what kinds of AI technologies are we referring to in this context? Here, we refer specifically to those technologies either derived from or powered by artificial intelligence. These include, for example, advanced AI customer service representatives that rely on natural language processing (NLP), AI-assisted medical imaging diagnostics that utilize machine learning, and autonomous robots that integrate robotics with AI's complex problem-solving capabilities. As these examples illustrate, AI technologies are being leveraged to enhance and augment existing occupations. This adaptability is one of AI's defining features: it can be embedded across a wide range of job functions. In this section, we will briefly examine several industries and occupational domains where AI-driven automation is already being adopted. Detailed examples of such applications are provided below.

Services

The service sector is one of the most labor-intensive industries. Typical roles in this sector include retail associates, food service workers, customer representatives, bartenders, baristas, and other public-facing occupations. AI-powered robotics are increasingly capable of replacing portions of these roles. More advanced robotic systems with agile mobility and intelligent processing capabilities are beginning to perform a broader range of service-related tasks—for instance, robotic bartenders, chefs, or even massage robots. Notably, a specialized subfield called embodied artificial intelligence seeks to develop AI systems with physical embodiments that can interact with the material world. Embodied AI holds significant promise for transforming labor demands in the service sector.

Manufacturing

The manufacturing industry is another sector heavily reliant on human labor. It encompasses a broad range of production activities, including electronics, textiles, automobiles, pharmaceuticals, biomedical devices, and chemical goods. As discussed in the historical review of industrial revolutions, manufacturing has long been a labor-dense sector. However, the shift from manual labor to automated machinery accelerated with the development and implementation of advanced semiconductors in industrial automation. AI's role in this transition is already visible, particularly in the context of "advanced manufacturing" or the emergence of the "smart factory." These shifts build on the foundation of traditional automation and deepen its capabilities through machine learning, predictive maintenance, and real-time adaptive control systems (Edlich et al., 2019).

Transportation and Logistics

Transportation and logistics now serve as foundational components of modern life. This includes air travel, rail systems, road-based transit, and the logistics sector, which underpins global supply chains and domestic delivery networks. According to a report by the McKinsey Global Institute, transportation and logistics represent the second-largest potential market for AI applications (Edlich et al., 2019). One prominent example of AI integration in this sector is AI-driven logistics optimization, which can significantly affect labor roles such as route planning, dispatch coordination, and supply chain management. These systems not only enhance operational efficiency but also reshape the responsibilities and demand for certain workforce roles.

The examples above illustrate how AI has impacted specific industries and their associated occupations. However, the influence of AI is not limited to disruption or a sharp reduction in available job positions. Recent research suggests that AI may instead signal a reform of existing industries. Given the nature of this reform and the reallocation of roles, the outcome involves not only layoffs but also the creation of new opportunities. These complex dynamics will be explored further in the following sections. Table 18.1 summarizes historical technological revolutions and their relationship to labor market fluctuations.

1.2 Job Polarization and Labor Segmentation

In the previous section, we explored the historical trajectory of labor market disruptions and the emerging influence of AI technologies on today's employment landscape. In this section, we turn to two increasingly visible phenomena that stem from the growing prevalence of automation and artificial intelligence in workplaces: job

Table 18.1 Historical technological revolutions and labor market disruption (Szeszák et al., 2025)^a

Revolution	Time period	Key technologies	Labor market disruption
First industrial revolution	Early 1800s	Steam power, mechanized textile production, water power	Marked by the transition from hand production to machinery. Labor-intensive industries like textiles were transformed by steam-powered mechanization. The shift drove population migration from countryside to cities, initiating the transformation from agrarian to industrial economies.
Second industrial revolution	Late 1800s—Early 1900s	Electricity, assembly line, telephone, electric lighting, radio	Characterized by the generalization of electricity and communication technologies. Invention of the assembly line boosted factory efficiency. Urbanization accelerated, and labor demands continued to rise due to mass production and technological advances in daily life.
Third industrial revolution	1950s—Early 2000s	Programmable automation, computers, semiconductors, internet	Initiated the shift to the digital era. Fully automated production processes emerged. New job categories like software engineers were created, while existing roles were restructured or disappeared. Labor markets adapted to growing reliance on digital tools and personal computing.
Fourth industrial revolution	2000s—Present	AI, robotics, IoT, AGI, cloud computing, edge computing	Rooted in digital technologies, this phase is reshaping all aspects of work. AI is seen as disruptive, large-scale, and workflow-transforming. Its ability to replace human labor across sectors may lead to rapid, large-scale unemployment and demands for urgent reform.

^aAdapted from Szeszák et al. (2025, Fig. 1). Some modifications were made. <https://doi.org/10.3390/soc15040088>, licensed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>)

polarization and labor market segmentation. These structural changes are not only reshaping occupational distributions but are also prompting serious concerns around employment stability, inequality, and long-term social cohesion.

Job polarization refers to the simultaneous expansion of high-skill and low-skill occupations, accompanied by a marked decline in middle-skill jobs. In other words, while demand for top-tier professionals and basic service workers continues to rise, traditionally stable mid-level jobs are diminishing (Goos et al., 2014). This trend reflects more profound transformations in work structures, in which occupations requiring moderate cognitive or manual skills—once considered the backbone of the modern economy—are increasingly susceptible to automation and displacement.

Closely related is the concept of labor segmentation, which describes the growing separation of the labor force into distinct clusters or sub-markets. These

sub-markets often differ significantly in terms of job security, wage levels, benefits, and advancement opportunities. Notably, movement between these segments is usually restricted, with structural barriers preventing many workers from accessing better opportunities. In academic discourse, this phenomenon is often referred to as the “dual labor market” model, which differentiates between the primary labor market—characterized by full-time, well-paid, and secure jobs—and the secondary labor market, which comprises part-time, temporary, or unstable jobs with lower compensation and limited protections (Piore, 2014).

One of the primary drivers of job polarization today is the automation of routine tasks, where AI has shown particular strength. Research suggests that AI systems excel at performing repetitive, rule-based tasks—such as data entry, administrative processing, and manufacturing assembly-line work. In contrast, non-routine cognitive tasks, which require creativity, adaptability, or interpersonal engagement—such as education, healthcare, or social work—remain much more resistant to full automation in the short term (Acemoglu & Restrepo, 2018). As a result, the workforce is being pushed toward a bifurcated structure: high-skill roles that complement AI flourish. At the same time, low-skill service jobs persist out of necessity, and many middle-skill jobs face increasing obsolescence.

These structural changes also expose and exacerbate regional and demographic disparities. High-skilled workers in urban centers and developed regions often benefit from the productivity gains and income boosts enabled by AI adoption. In contrast, workers in rural or economically underdeveloped areas face disproportionate risks of job loss and exclusion from digital transformation. Age, educational attainment, and gender further influence how different groups experience and navigate the impacts of AI.

To address these inequities, researchers and policymakers increasingly advocate for robust reskilling and upskilling programs (Li, 2024). These initiatives must go beyond technical training to include social and psychological support—areas where social workers can play a transformative role (Table 18.2). By participating in policy advocacy, providing career transition services, and empowering underserved

Table 18.2 Labor tasks by skill level and automation risk

Skill level	Typical tasks	Automation risk	Labor market segment
High-skill	Creative decision-making, unstructured communication, professional services (e.g., education, nursing)	Low —Hard to replace due to complexity, creativity, and interpersonal nature	Primary labor market (stable jobs, high salary, benefits)
Middle-skill	Routine office tasks, data entry, assembly-line work, administrative support	High —Highly automatable due to rule-based, repetitive characteristics	Shrinking segment, facing systematic reduction
Low-skill	Manual labor, basic services (e.g., cleaning, food service, delivery)	Moderate —Some tasks automatable, others still require physical presence and adaptability	Secondary labor market (part-time, temporary, low wage)

populations, social workers can help mitigate the disruptive effects of AI while fostering greater equity and social inclusion in an evolving labor market.

1.3 Case Studies

The impact of AI technologies on the labor market is reflected not only in systematic restructuring of work but also varies significantly across industries. In this section, we examine specific examples from the manufacturing sector, the retail and service industries, and the healthcare field to explore how AI technologies are transforming work processes and redistributing labor.

1.3.1 Case 1: Manufacturing Industry

In the manufacturing sector, AI technologies have primarily been applied to the operation and maintenance of automation systems. For example, industrial robots are widely used to perform highly repetitive, precision-based tasks on assembly lines. AI-driven smart maintenance systems improve operational efficiency by increasing equipment availability and minimizing malfunction-related interruptions (Chiacchio et al., 2018). These intelligent transformations have significantly enhanced manufacturing productivity. However, they have also resulted in layoffs among middle-skilled workers. Traditional roles in factories—such as assembly workers and inspectors—are gradually becoming obsolete. In response to these challenges, initiatives like Germany’s “Industry 4.0” not only promote the development of smart manufacturing but also emphasize the importance of vocational education. These policies encourage skill development in digital tools and advocate for new models of “human-machine collaboration.”

1.3.2 Case 2: Retail and Service Industries

The retail and service industries are at the forefront of labor displacement driven by automation. Automated technologies such as self-checkout systems and vending machines are becoming increasingly common in retail environments. At the same time, businesses are adopting AI systems to reduce human resource costs and continuously optimize customer service protocols. Although these technologies bring benefits to companies, they have also reduced labor demand for front-line workers. For instance, in chain supermarkets and fast-food restaurants, traditional cashier and server roles are being phased out in favor of automated systems such as self-order kiosks. Simultaneously, the spread of self-service machines has created new occupations, such as systems operations engineers. These emerging roles typically require higher skill levels, posing additional challenges for less experienced or non-traditional workers.

1.3.3 Case 3: Healthcare Industry

The healthcare industry presents a unique case in which AI brings both enhancement and displacement. Technologies such as AI-assisted diagnostic imaging systems are increasingly being used to support medical image analysis and clinical decision-making. These tools have significantly improved physician efficiency and the overall quality of healthcare delivery. However, at the same time, low-skill roles—such as electronic health record (EHR) recorders, medical coders, insurance reimbursement assistants, and clinical administrators—are gradually being replaced by AI-powered automation software. These changes reduce reliance on backend labor and raise concerns about data security, privacy, and ethical AI governance (Topol, 2019).

Table 18.3 summarizes the aforementioned cases. To address the challenges brought by AI implementation, governments and enterprises have developed a range of integrative strategies. For instance, several European Union member states have introduced worker-centered skill transformation and upskilling programs that promote internal training and workforce reallocation within the same organizations (Arntz et al., 2016). Social workers play a critical role in this process: not only can they provide counseling services to laid-off or affected workers, but they can also actively engage in policy advocacy and contribute to human-centered market development by building a more equitable and inclusive workplace and technological transition.

Table 18.3 Sectoral case summary: AI impact, response, and outcomes

Sector	AI application	Observed labor impact	Response and outcomes
Manufacturing	Industrial robots on assembly lines; smart maintenance systems	Lay-offs of middle-skilled workers; shrinking demand for assembly workers and inspectors	“Industrial 4.0” promotes vocational training and human-machine cooperation mechanisms (Chiacchio et al., 2018)
Retail & service	Self-checkout machines, vending machines, self-order kiosks; AI-enabled customer service	Reduced demand for front-line workers; replacement of cashiers and servers; emergence of new tech maintenance roles	Emergence of new occupations with higher skill requirements; challenges for less experienced and nontraditional workers
Healthcare	AI-assisted diagnostic systems for imaging and clinical support; automated backend documentation	Enhancement of physician efficiency; job loss for recorders, coders, administrators; concerns over data and AI ethics	Controversy over privacy and data protection; push for balanced automation and ethical governance (Topol, 2019)

2 Labor Rights and Algorithmic Workplaces

As AI technologies continue to permeate the workplace, an increasing number of work procedures and requirements are being delegated to data-driven algorithmic systems. Algorithms are no longer merely tools for decision-making; they have become central drivers of task management and work allocation across industries. These changes are significantly impacting the structure of workers' rights, their day-to-day experiences, and long-term career trajectories. In this chapter, we will examine the challenges that algorithmic workplaces pose to workers by focusing on three key aspects: first, the erosion of worker autonomy due to algorithmic management; second, the intensification of workplace surveillance and the related concerns about data privacy; and third, the presence of algorithmic bias in human resources processes such as recruitment and promotion. This chapter aims to explore how labor rights are being restructured in the age of technological transformation and to discuss potential strategies—such as policy reform, social work interventions, and technological oversight—to promote fairness and justice in AI-mediated work environments.

2.1 *Worker Autonomy*

The emergence of AI technologies has led to a significant expansion in the adoption of algorithmic management systems across a wide range of workplaces. Algorithmic management refers to end-to-end digitalized management processes powered by big data analytics and automated decision-making systems. These systems oversee task allocation, workflow organization, performance evaluation, scheduling, and even income distribution for workers. Such data-driven management approaches offer substantial advantages for enhancing operational efficiency and reducing costs. They have been widely implemented across sectors such as e-commerce, logistics, retail, manufacturing, and more (Jarrahi et al., 2021).

However, alongside these gains in efficiency, algorithmic management systems have substantially eroded front-line workers' autonomy. In traditional settings, human administrators were responsible for most decisions and communication. Today, those functions have largely been replaced by opaque rating mechanisms, performance indices, and algorithmic recommendations. It has become increasingly complex for workers to understand how decisions about their tasks are made or to question the fairness of those decisions (Kellogg et al., 2020). This issue is particularly prominent in platform-based economies such as ridesharing and food delivery services, where workers are technically classified as “independent operators” but are tightly constrained by platforms' algorithmic management systems.

More importantly, the rise of algorithmic management introduces significant ethical concerns. These systems make decisions based solely on data and historical

patterns, without incorporating empathy, context, or ethical reasoning. For instance, algorithms may reinforce systemic biases by assigning unfair weights or relying on historically biased datasets, resulting in discriminatory outcomes against specific groups of workers. Additionally, algorithmic decisions may fail to account for personal emergencies or exceptional circumstances, leading to unjust penalties or performance assessments. Since many of these systems are designed to minimize human oversight, they can further marginalize non-standard workers, such as those who are pregnant or living with physical or mental disabilities, exacerbating inequities in the workplace.

2.2 *Surveillance and Data Privacy*

Workplace surveillance has undergone a profound transformation due to advances in AI and sophisticated image processing. In the past, workplace monitoring primarily relied on human oversight. Today, it has gradually evolved into automated systems powered by intelligent algorithms and sensor technologies. Enterprises now employ facial recognition cameras, real-time location-tracking systems, device activity-monitoring tools, voice recognition systems, and other surveillance technologies to monitor employees' movements, productivity, and even their communication content (Ajunwa et al., 2017).

From a management perspective, these technologies facilitate more timely and quantifiable oversight, enabling firms to evaluate employee performance, monitor work status, and support cost control and operational efficiency. However, these benefits come with significant concerns. The legality and ethical implications of pervasive surveillance have sparked widespread controversy. The growing reliance on continuous monitoring raises critical questions about privacy infringement, informed consent, and the potential erosion of workers' dignity and autonomy in the modern workplace (Table 18.4).

Some scholars have pointed out that beyond improving efficiency and enabling micromanagement, workplace surveillance systems based on behavioral data collection are no longer merely management tools—they represent a form of “Surveillance Capitalism,” in which individual privacy is sacrificed to concentrate corporate power and maximize profit (Zuboff, 2019). When legal boundaries remain ambiguous, surveillance practices can more easily overstep legitimate oversight and infringe upon essential privacy rights and human dignity.

Even more far-reaching are the psychological and social consequences of such pervasive, 360-degree surveillance. Research has shown that workers subjected to high-pressure, continuous monitoring environments over the long term are more likely to experience anxiety, depression, burnout, and other emotional disorders (Moore et al., 2018). These adverse outcomes can lead to disengagement and detachment from work. Employees may lose motivation, experience a diminished sense of autonomy, and develop passive or defensive attitudes, which in turn contribute to increased tension and deteriorating workplace relationships.

Table 18.4 Types of workplace Surveillance and risk dimensions (Ball, 2021)^a

Surveillance type	Technical features	Primary purpose	Associated risks
Face-recognition cameras	Image-based identification powered by AI algorithms	Monitor presence and behavior	Potential violation of privacy rights; controversial in legitimacy and ethical boundaries
Position tracking systems	Location sensors detecting real-time employee movements	Monitor mobility and working hours	Possible over-monitoring and lack of transparency
Working device monitoring systems	Record keyboard, mouse, application usage, system activity	Evaluate efficiency and workflow discipline	Data overcollection; psychological burden due to constant behavioral measurement
Voice recognition systems	Analyze audio inputs to track communication and response times	Improve response efficiency and control protocols	Ethical concerns about monitoring communication contents; employee trust erosion
Integrated AI-based surveillance	Combine multiple sensors and algorithms for 24/7 full-range tracking	Real-time performance metrics and cost control	Risk of psychological pressure, emotional fatigue, anxiety, burnout, and workplace disengagement

^aAdapted from Ball (2021), Chap. 2: New workplace surveillance practices. Some modifications were made. <https://publications.jrc.ec.europa.eu/repository/handle/JRC125716>, licensed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>)

2.3 Bias and Fairness

In the previous section, we explored how algorithmic management can undermine workers’ autonomy and how workplace surveillance can violate employee privacy and harm emotional well-being. Although AI technologies are often portrayed as objective and neutral, their decision-making processes are inherently rooted in existing data and social structures. As a result, algorithmic bias is embedded within systems that may appear fair on the surface.

Barocas and Selbst (2016) examined algorithmic models adopted by numerous companies for human resource functions such as recruitment, screening, and performance evaluation. They found that these models are often trained on historical employee data that already contains various forms of bias—such as information bias, selection bias, and demographic inequalities in gender, ethnicity, and education. When such biased data are used to train AI-powered algorithms, the resulting systems can perpetuate and even amplify discrimination. For instance, studies have shown that some automated hiring systems tend to exclude nontraditional job

seekers—such as women applying for technical roles or minority candidates—because they are underrepresented in historical datasets (Raghavan et al., 2020).

To mitigate these biases, scholars have emphasized the importance of transparency and ethical auditing in AI systems that affect workers' rights and livelihoods. Transparency measures may include disclosing the sources of training data and the logic behind algorithmic decisions. Furthermore, implementing fairness evaluation procedures involving multidisciplinary reviewers and introducing human oversight at critical decision points can help ensure accountability.

Social workers can play a vital role in addressing these issues by educating workers about algorithmic bias and their workplace rights (Eubanks, 2018). They may also advocate for ethical AI use by collaborating with government agencies and nonprofit organizations to encourage businesses to incorporate social responsibility as a core consideration when adopting new technologies.

3 Ethical and Societal Implications

In the previous sections, we examined the impact of AI on labor market structures, workers' rights, and the workplace environment. Beyond these direct effects, AI also generates broader influences on society and ethics. In this chapter, we will further explore the ethical and societal implications of AI, focusing on three key areas: economic inequality, responsible AI development, and regulatory and policy interventions.

3.1 *Economic Inequality*

AI technologies have been regarded as powerful tools for promoting efficiency and increasing profits. However, they also risk widening existing disparities in wealth and opportunity. Research has shown that AI technologies are primarily controlled by major enterprises with advantages in capital, data, and technical expertise (Korinek & Stiglitz, 2021). These benefits tend to accrue first to high-skill workers and capital owners, further empowering already advantaged populations as active users and participants. In contrast, low- and middle-skill workers, along with marginalized populations, resort to being the passive target of AI-based information and organizational decisions, thereby facing significantly greater risks of job loss. The unequal distribution of access and capacity to implement AI further exacerbates technological divides, acting as a powerful amplifier of socioeconomic inequality.

3.2 *Responsible AI Development*

To address the structural imbalances created by AI technologies, both academic and policy communities have increasingly emphasized the need for “Responsible AI.” Responsible AI calls for embedding essential ethical values and public interest principles into both technological development and business applications. Researchers such as Jobin et al. have identified several core design principles for responsible AI systems: explainability, fairness, accountability, transparency, and safety (Jobin et al., 2019). These principles are recognized as critical safeguards against AI systems that may harm vulnerable populations, infringe on human rights, or undermine public trust. Furthermore, scholars argue that the governance of AI should not be confined to technical regulation alone but must also address how these technologies are socially embedded and interact with broader societal structures. The development of responsible AI should also be expanded to promote humanistic values in the workplace, so that AI for employee engagement can serve as the foundation upon which enterprise productivity can rest (Hong, 2024). The control mechanism of AI for business interests and efficiency can be offset by AI’s role in advancing an inclusive workplace and culture development. A fair AI system should not only be technically controllable but also socially accountable (Crawford & Calo, 2016).

3.3 *Regulatory and Policy Interventions*

To effectively manage the societal risks posed by AI, a variety of governance frameworks and regulatory strategies have been proposed across different regions. The European Union, for example, introduced one of the world’s first comprehensive regulatory efforts through the “Artificial Intelligence Act” (Regulamento da União Europeia (UE), 2024). This act employs a tiered, risk-based framework that reinforces approval and accountability requirements for high-risk AI systems. In contrast, the United States has taken a more market-oriented approach, emphasized innovation, and limited regulatory intervention, thus avoiding a unified legislative framework. In response to these divergent strategies, scholars and policymakers have advocated for a global regulatory approach grounded in human-centered values. Such a framework should prioritize: (1) the development of standardized international regulations for data usage and platform accountability; (2) protections against systematic bias and discrimination, especially for vulnerable populations; and (3) the inclusion of governments and civil society in auditing and screening AI systems during the design phase to prevent large-scale systemic risks (Table 18.5).

Table 18.5 Key points on ethical and societal implications of AI

Section	Main points	Key references	Policy directions/ suggestions
Economic inequality	AI increases efficiency but widens wealth and opportunity gaps. Benefits go to high-skill workers and capitalists; low/mid-skill and marginalized groups face job loss	Korinek and Stiglitz (2021)—AI under control of large enterprises with capital, data, and technology advantages; amplifies socioeconomic disparity	Address technology-driven inequality; mitigate impact on vulnerable populations
Responsible AI development	Embed ethics into AI via explainability, fairness, accountability, transparency, and safety. Focus on social impact, not just technology	Jobin et al. (2019)—Design principles to protect vulnerable groups and human rights; Crawford and Calo (2016)—Focus on social embeddedness of AI	Ensure AI is not only technically controllable but socially accountable
Regulatory and policy interventions	EU adopts risk-based AI act; US favors market-led approach. Global push for human-centered regulation	EU AI Act (2024); USA market-oriented approach	(1) Standardized data regulation and platform accountability; (2) Prevent systematic bias/discrimination; (3) Encourage ethical review during algorithm design

4 Social Work and AI: Mitigation and Advocacy

AI technologies have garnered increasing attention due to their profound impact on societal structures, including labor market disruptions and ethical concerns. In this context, social support systems—particularly social work—become even more essential alongside technical interventions. Social workers, grounded in justice and human-centered values, play a critical role in addressing current challenges posed by AI and in advocating for and offering equitable, future-oriented solutions.

4.1 Policy Advocacy and Practice

In previous sections, we discussed the impacts of AI technologies on economic structures, technical design, and ethical governance. We also highlighted the associated risks and broader societal influences. However, technical optimization and policy development alone are not sufficient to achieve inclusive technological transformation. Social workers can become a vital force in promoting justice and equitable AI implementation by advocating for socially beneficial policies and practices.

Social workers possess long-standing practical experience in engaging with policymaking processes and addressing structural disparities. They can offer professional, human-centered perspectives in debates concerning AI technologies. Reamer (2023) emphasizes that social workers should participate in shaping governance and public policy. Specifically, they should advocate for the inclusion of social justice, data ethics, and the protection of vulnerable populations within AI-related policies. Such advocacy can take the form of conducting research, delivering conference presentations, participating in community engagement efforts, and translating social observations into policy recommendations with systemic influence.

Social workers should also pay special attention to marginalized populations affected by the AI revolution—such as low-skill workers, older adults, immigrant workers, and people with disabilities. As Wronka (2022) argues, the ethical foundation of social work obliges practitioners to advocate for the rights of minority and vulnerable groups. In the context of labor market challenges caused by automation, algorithmic discrimination, and data exploitation, social workers should actively engage with lawmakers, educational institutions, and nonprofit organizations to ensure these populations are not left behind. Social workers can also directly participate in the practice of developing alternative or competing human-centered algorithms that could generate dynamic information (1) offsetting disadvantages experienced by vulnerable populations in the world of AI and (2) feeding into other AI algorithmic functions as essential data for the larger system level to benefit (Hong, 2024). By having more algorithms at the system level rely on quality human-centered data from marginalized populations, the adverse effects in the labor market caused by automation, algorithmic discrimination, and data exploitation can be minimized.

Equally important, social work practitioners can serve as a bridge between the general public and government institutions. They can help ensure that AI governance reflects public interest by participating in policy discussions, submitting policy memos, or serving as expert consultants on policymaking committees. For example, social workers could recommend developing algorithmic transparency standards, implementing labor data privacy regulations, and participating in AI ethics review boards to safeguard the public good.

In summary, AI governance should not be the exclusive domain of technologists and legislators. Social workers must actively contribute to this mission by leveraging their experience in community engagement and direct service provision. Their involvement in policy advocacy and practice is essential in guiding AI development toward fairness, inclusivity, and responsibility—values that reflect the core mission of social work and promote social justice in an increasingly digital society.

4.2 Supporting Displaced Workers

Traditional labor market structures are undergoing systematic restructuring, driven by the rapid development of artificial intelligence technologies. These unforeseen transformations have led to unprecedented displacement, particularly among

middle- and low-skilled workers. Many individuals are now facing the threat of job loss, career disruption, and forced repositioning. These challenges impact not only their emotional well-being but also their ability to adapt socially. At this critical juncture, social work practitioners should serve as first responders, offering direct services and support to affected individuals.

Clients laid off due to AI-related disruptions often experience emotional symptoms similar to those observed in the broader unemployed population. These may include anxiety, denial, self-blame, and helplessness. In more severe cases, individuals may develop symptoms of post-traumatic stress or depression (Dziegielewski, 2013). Drawing on their expertise in psychological and psychosocial support, social workers can provide both individual and group counseling to help unemployed individuals process their feelings, recognize external factors driving these changes, rebuild confidence, and gradually adjust to a new life rhythm. At the same time, social workers can establish “unemployment mutual help clubs” to foster mutual sharing and peer support. For example, an evidence-based social work intervention called Transforming Impossible into Possible (TIP) uses the psychological self-sufficiency (PSS) theory to shift perceived employment barriers into employment hope, supporting individuals who are voluntarily or involuntarily disconnected from the labor market (Hong, 2013, 2016). These community-based interactions can help reduce social isolation and self-blame while enhancing social belonging and psychological resilience.

In addition to providing emotional support, social workers should take a proactive role in position transition programs and digital skills retraining initiatives. The National Association of Social Workers (NASW, 2024) emphasizes that social workers can provide critical services in these areas, including resource referrals, capability assessments, action planning, and support for career reconstruction. Using the same example from above, the PSS framework offers a system for evaluating individualized goal achievement pathways based on perceived employment barriers and employment hope over time (Hong, 2014; Hong et al., 2012, 2014). TIP intervention is designed to strengthen the PSS process for overcoming barriers to hope and achieving disconnected workers’ employment goals (Hong, 2016). This involves helping clients identify their core strengths, recognize their potential, and develop realistic, goal-oriented learning plans. Through collaboration with vocational schools, higher education institutions, and nonprofit organizations, social workers can refer clients to appropriate vocational education programs. They can also provide ongoing support, including financial aid application assistance, career counseling, and post-placement support. Such partnership-based collaborative interventions facilitate a shift from passive unemployment to proactive re-engagement with the workforce.

In summary, labor market transformation driven by the AI revolution should not be viewed solely as a structural challenge, but also as a critical opportunity for social work innovation (Table 18.6). By providing emotional support, conducting individualized assessments, referring to job resources, supporting skill development, and fostering community-based networks, social workers can help displaced individuals transition from victims of disruption to adaptive, empowered contributors to the evolving economy.

Table 18.6 Reskilling and support program summary

Action	Brief description	Key resources	Key stakeholders
1. Emotional counseling	Provide individual or group counseling to help process anxiety, denial, self-blame, and rebuild stability.	Counseling rooms, trauma-informed care tools, trained staff	Social workers, mental health professionals
2. Peer support facilitation	Organize mutual help clubs to reduce isolation, foster peer interaction, and build social belonging.	Community centers, facilitation guides, peer leaders	Unemployed clients, community organizers
3. Capability assessment	Support identification of strengths, transferable skills, and adaptability for future planning.	Assessment tools, digital skills frameworks	Social workers, career coaches
4. Learning path planning	Design realistic and goal-oriented reskilling or education plans tailored to client potential.	Education catalogs, skill gap analysis tools	Vocational schools, adult education providers
5. Resource Referral & Ongoing Support	Connect to training programs, assist with financial aid, and offer career navigation support.	Training referrals, funding applications, job navigation tools	NGOs, workforce agencies, financial aid officers

4.3 Empowerment and Education

While the previous section focused on practical short-term interventions, empowerment and education offer broader, long-term strategies for addressing the challenges posed by AI-driven labor market disruption. Empowerment can be fostered through education that promotes digital literacy and strengthens awareness of workplace rights (Finn, 2020).

Social workers can play a key role in this process by organizing workshops and community presentations that help individuals understand how algorithmic systems function and how to identify and respond to technological unfairness in employment contexts. These efforts enable communities to become more informed and resilient in the face of rapid digital transformation.

In addition, social workers can support labor unions and worker-based organizations in rebuilding collective communication mechanisms. This is particularly vital for platform-based workers, who often lack traditional representation. By helping these workers establish mutual support networks and communication channels, social workers contribute to a stronger worker voice and shared problem-solving (Scholz, 2017).

In summary, empowerment and education are essential dimensions of social work practice. They equip individuals and communities with the tools to navigate and challenge the structural inequalities arising from AI adoption, while advancing broader labor market transformation.

5 Future Directions and Recommendations

5.1 *Characteristic Shifting of Social Work in AI Labor Market*

As AI becomes more deeply embedded into workplace structures, the practice model of social work faces unprecedented challenges—alongside valuable opportunities. Traditionally, social work has focused primarily on providing individualized services, including incident-based interventions and resource referrals, with relatively less attention to group-level or systemic advocacy. However, in the current context, where AI is reshaping the labor market and workplace governance, social workers must expand their roles to engage more actively in structural systems and contribute their expertise to building ethically grounded smart workplaces.

As discussed in previous sections, social work practitioners should proactively participate in algorithmic systems review processes, particularly for AI systems used in recruitment, performance evaluation, and platform-based task assignment—all of which are directly tied to workers' income and employment stability. Social workers can advocate for the integration of transparent, explainable algorithms and support the development of fair mechanisms that include specialized indexes, such as a “vulnerable population impact assessment.” One such example presented earlier is the overtime PSS measures of perceived employment barriers and employment hope that can be incorporated into the algorithm for employee success on the job (Hong, 2024). Introducing such considerations early in the design phase can help mitigate biases and discriminatory outcomes that may be embedded in the historical data used to train AI models.

Moreover, social workers should broaden their professional boundaries by collaborating within interdisciplinary research teams that include data ethicists, AI engineers, policy analysts, business leaders, human resource specialists, and lawmakers. These interdisciplinary teams can collaborate to develop human-centered governance frameworks for AI-integrated workplaces. For example, social workers can help design and implement community-based support projects tailored to emerging labor arrangements such as part-time gig work, platform-based economies, and remote employment.

These support initiatives should offer not only material resources but also psychological resilience counseling and career navigation services. By strengthening workers' negotiation and bargaining capacity in algorithm-dominated work environments, social workers can help enhance civic participation and social protection in the era of intelligent labor systems.

5.2 Future Research and Interdisciplinary Cooperation

Although current studies have begun to explore the impact of AI technologies on the labor market, many critical questions remain insufficiently addressed. One key research priority is to understand AI’s influence on occupational identity, self-perception, and long-term emotional well-being. In particular, within nontraditional employment models—such as platform economies and flexible or precarious work arrangements—there is a lack of systematic assessment of how unstable job conditions and social isolation affect workers over time. These conditions may be captured by interdisciplinary collaborations to study various dimensions of employment barriers in both individual and structural contexts (Hong et al., 2021).

Furthermore, research is needed to explore how social work can be systematically embedded within AI governance systems. This includes going beyond individual counseling to contribute meaningfully to policy development, ethical algorithm review, and inclusive workflow design. Social workers must not only respond to the impacts of technology but also proactively shape the environments in which such technologies are deployed. This would also mean that social workers need to contribute to business decisions by proactively investing in AI algorithms for human-centered career development processes, from employee hiring and retention to career advancement, that will ultimately benefit productivity (Hong, 2024).

Emerging scholarship suggests that future AI governance will require deep and sustained collaboration between technological development and social science inquiry (Serena & Kneese, 2024). To foster this interdisciplinary integration, it is essential to establish virtual and institutional collaboration models—what might be called a “Social Work–AI–Policy” triad—that promotes the meaningful inclusion of social workers in shaping equitable AI-driven workplace transitions.

Social workers can also collaborate with institutions of higher education to co-develop academic programs such as “AI and Social Justice.” Such curricula can prepare a new generation of interdisciplinary scholars and practitioners who possess both ethical sensitivity and technical fluency in evolving technologies.

In short, the AI-driven transformation of the workplace is an irreversible trend (Table 18.7). This moment offers a rare and valuable opportunity for social work

Table 18.7 Summary of future directions and cooperation methods

Future direction	Key actions	Cooperation methods
1. Embed social work into AI governance systems	Expand social work roles to include policymaking, ethics review, and workflow design	Joint taskforces with tech developers, ethics boards, and public agencies
2. Build “social work—AI—Policymaking” collaborative workflows	Develop systematic collaboration models for just and equitable AI-driven workplace transitions	Establish cross-sector virtual labs with social workers, AI engineers, and public policymakers
3. Assess AI’s impact on occupational identity and well-being	Study the psychological and emotional effects of flexible/platform work and technological displacement	Interdisciplinary research involving social work, sociology, and occupational psychology

practitioners to solidify their role in the governance of future smart societies. By engaging in research, education, and policy, social workers can help lead the transition toward more just, inclusive, and sustainable digital labor systems.

6 Summary

To address the challenges posed by the AI-driven workplace revolution, social workers should proactively engage in interdisciplinary research, business engagement, and policy and community advocacy. By leveraging their expertise in counseling, organizational, community, and policy practice, they can support individuals in transition and help build a more inclusive, just, and sustainable workplace environment.

Competing Interests The authors have no conflicts of interest to declare that are relevant to the content of this chapter.

References

- Acemoglu, D., & Restrepo, P. (2018). Low-skill and high-skill automation. *Journal of Human Capital*, 12(2), 204–232. <https://doi.org/10.1086/697242>
- Ajunwa, I., Crawford, K., & Schultz, J. (2017). Limitless worker surveillance. *California Law Review*, 105, 735. <https://doi.org/10.15779/Z38BR8MF94>
- Arntz, M., Gregory, T., & Zierahn, U. (2016, May 13). *SeloVerde uses geospatial big data and AI/ML to monitor deforestation in supply chains, powered by AWS*. https://www.oecd.org/en/publications/the-risk-of-automation-for-jobs-in-oecd-countries_5jlz9h56dvq7-en.html
- Ball, K. (2021). *Electronic monitoring and surveillance in the workplace. Literature review and policy recommendations*. Publications Office of the European Union. <https://doi.org/10.2760/5137>. ISBN 978-92-76-43340-8.
- Barocas, S., & Selbst, A. D. (2016). Big data's disparate impact. *California Law Review*, 104, 671.
- Chiacchio, F., Petropoulos, G., & Pichler, D. (2018). *The impact of industrial robots on EU employment and wages: A local labour market approach (working paper 2018/02)*. Bruegel Working Paper. <https://www.econstor.eu/handle/10419/207001>
- Crawford, K., & Calo, R. (2016). There is a blind spot in AI research. *Nature*, 538(7625), 311–313. <https://doi.org/10.1038/538311a>
- Dziegielewska, S. F. (2013). *DSM-IV-TR in action: Includes DSM-5 update chapter*. Wiley.
- Edlich, A., Phalin, G., Jogani, R., & Kaniyar, S. (2019). *Driving impact at scale from automation and AI*.
- Eubanks, V. (2018). *Automating inequality: How high-tech tools profile, police, and punish the poor*. St. Martin's Press.
- Finn, J. L. (2020). *Just practice: A social justice approach to social work*. Oxford University Press.
- Goos, M., Manning, A., & Salomons, A. (2014). Explaining job polarization: Routine-biased technological change and offshoring. *American Economic Review*, 104(8), 2509–2526. <https://doi.org/10.1257/aer.104.8.2509>
- Groumpos, P. P. (2021). A critical historical and scientific overview of all industrial revolutions. *IFAC-PapersOnLine*, 54(13), 464–471. <https://doi.org/10.1016/j.ifacol.2021.10.492>

- Hong, P. Y. P. (2013). Toward a client-centered benchmark for self-sufficiency: Evaluating the 'process' of becoming job ready. *Journal of Community Practice*, 21, 356–378.
- Hong, P. Y. P. (2014). Employment hope: A path to empowering disconnected workers. In E. J. Clark & E. F. Hoffer (Eds.), *Hope matters: The power of social work* (pp. 143–148). NASW Press.
- Hong, P. Y. P. (2016). Transforming impossible into possible (TIP): A bottom-up practice in workforce development for low-income jobseekers. *Environment and Social Psychology*, 1, 93–104.
- Hong, P. Y. P. (2024). Transforming opportunities into productivity (TOP): A new social development model for social transformation. *Social Development Issues*, 46, 88–108.
- Hong, P. Y. P., Polanin, J. R., & Pigott, T. D. (2012). Validation of the employment hope scale: Measuring psychological self-sufficiency among low-income jobseekers. *Research on Social Work Practice*, 22, 323–332.
- Hong, P. Y. P., Polanin, J. R., Key, W., & Choi, S. (2014). Development of the perceived employment barrier scale (PEBS): Measuring psychological self-sufficiency. *Journal of Community Psychology*, 42, 689–706.
- Hong, P. Y. P., Gumz, E., Choi, S., Crawley, B., & Cho, J. (2021). Centering on structural and individual employment barriers for social and human development. *Social Development Issues*, 43, 29–53.
- Jarrah, M. H., Newlands, G., Lee, M. K., Wolf, C. T., Kinder, E., & Sutherland, W. (2021). Algorithmic management in a work context. *Big Data & Society*, 8(2), 20539517211020332. <https://doi.org/10.1177/20539517211020332>
- Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1(9), 389–399. <https://doi.org/10.1038/s42256-019-0088-2>
- Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals*, 14(1), 366–410. <https://doi.org/10.5465/annals.2018.0174>
- Korinek, A., & Stiglitz, J. E. (2021). *Artificial intelligence, globalization, and strategies for economic development* (working paper 28453). National Bureau of Economic Research. <https://doi.org/10.3386/w28453>
- Li, L. (2024). Reskilling and upskilling the future-ready workforce for industry 4.0 and beyond. *Information Systems Frontiers*, 26(5), 1697–1712. <https://doi.org/10.1007/s10796-022-10308-y>
- McLean, S., Read, G. J., Thompson, J., Baber, C., Stanton, N. A., & Salmon, P. M. (2023). The risks associated with artificial general intelligence: A systematic review. *Journal of Experimental & Theoretical Artificial Intelligence*, 35(5), 649–663. <https://doi.org/10.1080/0952813X.2021.1964003>
- Moore, P. V., Upchurch, M., & Whittaker, X. (2018). Humans and machines at work: Monitoring, surveillance and automation in contemporary capitalism. In P. V. Moore, M. Upchurch, & X. Whittaker (Eds.), *Humans and machines at work: Monitoring, surveillance and automation in contemporary capitalism* (pp. 1–16). Springer. https://doi.org/10.1007/978-3-319-58232-0_1
- NASW. (2024). *NASW practice standards & guidelines [guideline]*. NASW. <https://www.social-workers.org/Practice/Practice-Standards-Guidelines>
- Nguyen, Q. P., & Vo, D. H. (2022). Artificial intelligence and unemployment: An international evidence. *Structural Change and Economic Dynamics*, 63, 40–55. <https://doi.org/10.1016/j.strueco.2022.09.003>
- Piore, M. J. (2014). The dual labor market: Theory and implications. In *Social Stratification* (4th ed.). Routledge.
- Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020). Mitigating bias in algorithmic hiring: Evaluating claims and practices. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, pp. 469–481. <https://doi.org/10.1145/3351095.3372828>
- Reamer, F. G. (2023). Artificial intelligence in social work: Emerging ethical issues. *International Journal of Social Work Values and Ethics*, 20(2), 52–71. <https://doi.org/10.5552/10-020-205>
- Regulamento da União Europeia (UE). (2024). *EU Artificial Intelligence Act | Up-to-date developments and analyses of the EU AI Act*. <https://artificialintelligenceact.eu/>

- Scholz, T. (2017). *Überworked and underpaid: How workers are disrupting the digital economy*. Wiley.
- Serena, O., & Kneese, T. (2024, May). *AI governance needs sociotechnical expertise*. Data & Society. <https://datasociety.net/library/ai-governance-needs-sociotechnical-expertise/>
- Szeszák, B. M., Keréjártó, I. G., Soltész, L., Galambos, P., Szeszák, B. M., Keréjártó, I. G., Soltész, L., & Galambos, P. (2025). Industrial revolutions and automation: Tracing economic and social transformations of manufacturing. *Societies*, 15(4). <https://doi.org/10.3390/soc15040088>
- Topol, E. J. (2019). High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine*, 25(1), 44–56. <https://doi.org/10.1038/s41591-018-0300-7>
- Wronka, J. (2022). A human rights framework for integrated practice. In *Human rights and social justice*. Routledge.
- Zuboff, S. (2019). Surveillance capitalism and the challenge of collective action. *New Labor Forum*, 28(1), 10–29. <https://doi.org/10.1177/1095796018819461>

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

